

Labwork 2: Get to Know Your GPU

Device Information Report

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Overview

For this lab, I ran the provided program on my computer to check whether it could detect and support my GPU. The program found one CUDA device and printed its basic information.

Device Detection Summary

My computer has an **NVIDIA GeForce GTX 1650**. It was detected as Device 0 and marked as **SUPPORTED**. The summary also shows that 1 out of 1 detected devices is supported.

Detailed GPU Specifications

The information reported by the program is listed below:

- **Device Name:** NVIDIA GeForce GTX 1650
- **Compute Capability:** 7.5
- **Support Status:** SUPPORTED
- **PCI Device ID:** 0
- **PCI Bus ID:** 1
- **UUID:** GPU-86f77b16-96d1-d7bc-3e85-6ac8c390a63e
- **Watchdog:** Enabled
- **Compute Mode:** WDDM (Windows Display Driver Model)
- **FP32/FP64 Performance Ratio:** 32

Analysis

The GTX 1650 has compute capability 7.5 and uses NVIDIA's Turing architecture. This means it can run CUDA programs that support this compute capability. The FP32/FP64 performance ratio is 32, so the card is much faster at single-precision calculations than at double-precision calculations. This is normal for a consumer graphics card such as the GTX 1650.

Note: The last line says "No CUDA-capable GPU found," but this does not match the result above it. The program already found the GTX 1650 and reported that 1/1 devices are supported. Therefore, the final line is probably printed by mistake because of an error in the program's condition, not because the computer has no CUDA GPU.